

2.3 INDOOR AIR QUALITY

National Environment Protection Measures exist to achieve ambient air quality that allows for the adequate protection of human health and well-being. Solid fuel burning is associated with adverse health effects, including respiratory effects in adults.

A systematic review on solid fuel combustion exposure and respiratory health in adults in Europe, USA, Canada, Australia and New Zealand found that reducing solid fuel burning improves air quality and improves respiratory health (Guercio et. al., 2022). Similarly, the combustion of natural gas in homes for cooking and space heating purposes is linked to 12% of asthma related cases in Australia (Knibbs et al., 2018). This is a result of chemicals such as nitrogen oxides, carbon monoxide, and sulfur dioxide, as well as particulate matter (PM_{2.5}) and formaldehyde, all of which cause inflammation in airways which can result in asthma symptoms (Musgrave, 2020). Electrifying our homes will reduce these pollutants, improving our indoor and outdoor air quality, with significant health benefits for homes and workplaces.

This DCP chapter looks to promote human health through a reduction in polluting fuels and increased ventilation requirements.

Objectives

- (a) To ensure that ambient air quality levels as specified in the National Environment Protection Measure (Ambient Air Quality) are met for:
 - Carbon monoxide
 - Nitrogen dioxide
 - Ozone
 - Sulfur dioxide
 - Particulate matter (PM₁₀ and PM_{2.5})
- (b) To improve Indoor Air Quality (IAQ) levels in the built environment, specifically for:
 - Nitrogen oxides
 - PM
 - Volatile Organic Compounds (VOCs)
 - Poly Vinyl Chloride (PVC) and
 - Mould

Controls

- (a) All residential development must enable ventilation:
 - (i) Windows must be openable excluding windows that are for light ingress or privacy purposes.
 - (ii) Carpark ventilation required under Building Code of Australia clause F4.11 must also integrate CO monitoring and Variable Speed Drive motors.
- (b) Solid fuel heating and cooking systems are not permitted in any development.
- (c) Gas cooktops, gas ovens or gas internal space heating systems are not permitted in any residential development. Instead, electric systems should be installed and clearly marked on DA plans.